Looking Ahead in Goal-Oriented Dialogue

Preference-Tree (PTO) vs Group-Relative (GRPO) optimization of a small Motivational-Interviewing therapist

Exp3 — results, measurement validity, and what to decide

Update on the 26 July snapshot: the evaluation has since been re-run end-to-end under a second, independent judge.

Lior Baruch · Reichman University · meeting with Kfir Bar · 3 August 2026

Llama-3.2-1B therapist (bf16) · gpt-4o-mini simulated patient + oracle · Claude Haiku 4.5 held-out judge · 96 personas, persona-paired

AGENDA

Where things stand, and what I'd like to decide today

In hand

- **The main comparison is finished.** PTO and GRPO, 10 training iterations each under matched settings, fully scored on the whole battery over 96 fixed patient personas.
- **The measurement instrument is now measured, not assumed.** This is what's new since the email: the oracle's own repeatability, and a complete re-scoring of every conversation by a judge from a different model family that never played the patient.
- Everything is reproducible from one command — figures, tables and summaries regenerate under either grader, with seeded confidence intervals.

Open — the agenda

- **How to tell the story.** Method paper, look-ahead paper, or MI paper — the choice sets both the venue and the shape of the thesis chapter.
- **Whether to finish the look-ahead (K=5) arms**, which are paused on API budget and are the one comparison still open.
- **What else is worth running** — in particular human MI-coder validation, which is the one remaining validity gap that money cannot close.
- **Scope.** What goes in a paper, what stays in the thesis, and in what order they get written.

In one line: the PTO-vs-GRPO result is finished and now survives a grader that took no part in training. What is left to decide is how we frame it, and whether the look-ahead comparison is worth buying.

The starting point — the ICLR 2025 paper

Preference Tree Optimization: Enhancing Goal-Oriented Dialogue with Look-Ahead Simulations

Baruch, Butman, Bar, Friedman · ICLR 2025

- **The method.** At each therapist turn, branch several candidate replies; for each candidate simulate **K further turns** of the conversation; let the oracle score the resulting trajectory; keep the best and worst as a preference pair; update with DPO. Repeat, regenerating the tree from the improved model each round.

- **Why look ahead.** Scoring a reply on its own rewards openings that look good in isolation. Scoring the reply plus K simulated turns rewards openings that lead somewhere.

- **Setup.** Llama-2-7B therapist, GPT-3.5 as both simulated patient and oracle, 96 patient personas, $K \in \{0, 5\}$, 7 iterations, reward = mean(Q1, Q2).

- **Headline finding.** Every PTO model beat the untrained baseline on session satisfaction and working alliance, and **K = 5 gave higher and more stable scores than K = 0**.

What changed since the paper

	Exp1 — ICLR paper	Exp2	Exp3 — this meeting
Therapist	Llama-2-7B	Llama-3.2-1B, 4-bit	Llama-3.2-1B, bf16
Patient + oracle	GPT-3.5	gpt-4o-mini	gpt-4o-mini
Patient personas	cooperative	less cooperative	less cooperative
Oracle output	regex-parsed	JSON schema	JSON schema
Evaluation	Q1, Q2	6 questionnaires	8 metrics × 2 independent judges
Methods	PTO	PTO (4 oracles) + weak GRPO	PTO vs GRPO, matched
Iterations	7	7 per arm	10 per arm (K=0)

Each experiment is a fresh re-implementation — no data carries over between them.

One caveat worth stating up front — Exp2 and Exp3 absolute scores are not on the same axis.

Same therapist model, but Exp2 generated its conversations in 4-bit and Exp3 in bf16. 4-bit produces far more degenerate, looping therapist turns (≈9.5% of turns vs ≈0.3%), which the oracle floors — so Exp2's base model scores 2.38 and Exp3's scores 3.00 for the same model. The non-degenerate Exp2 subset scores ≈2.93. Compare within an experiment, not across.

SETUP

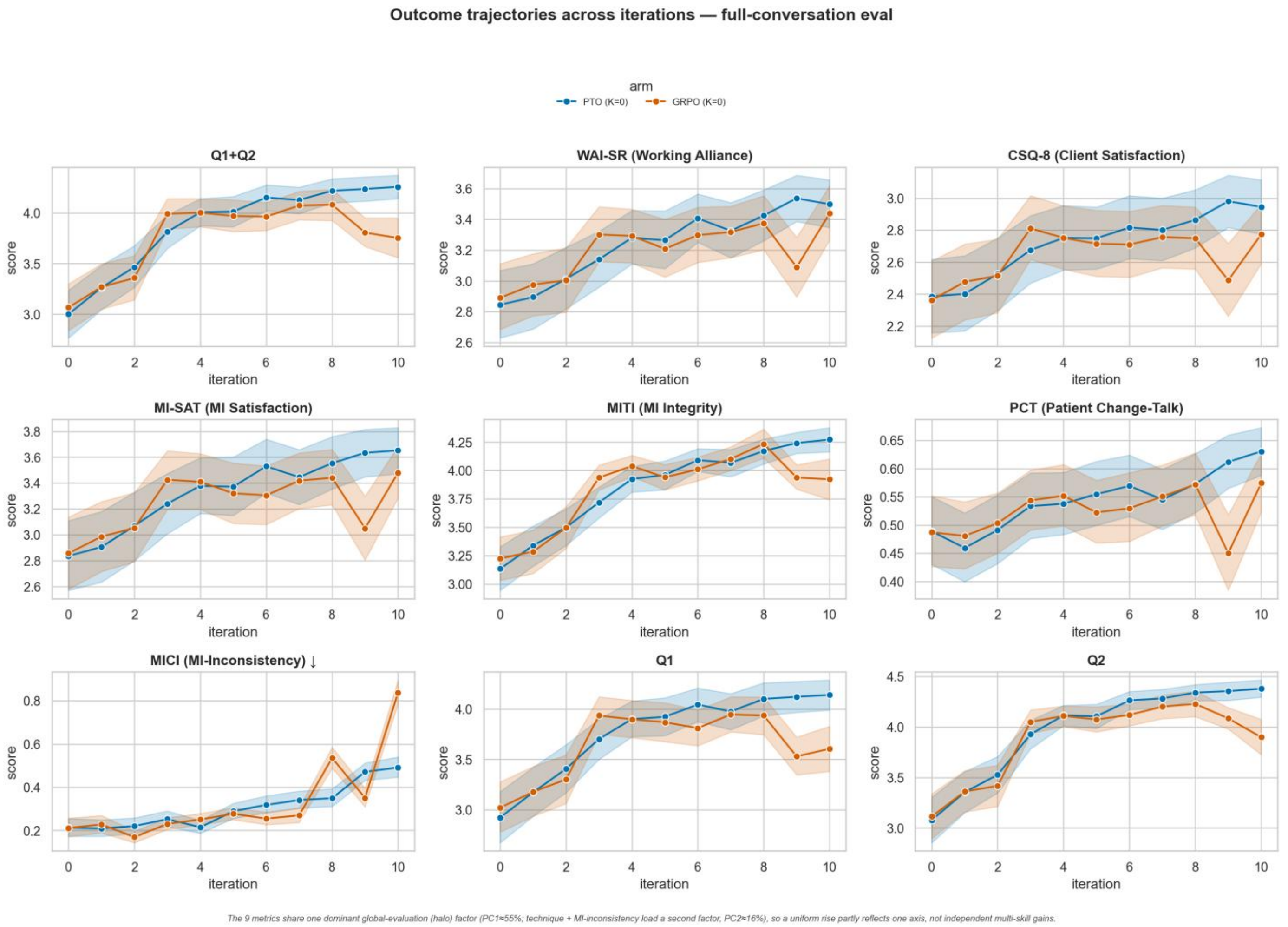
What was run in this experiment

- **Task.** A 1B therapist model converses with a simulated patient (96 fixed personas). A larger oracle model then grades the full conversation on validated MI questionnaires.
- **Training signal.** Oracle score on Q1+Q2 only. All other questionnaires are held out for evaluation.
- **Methods compared.** PTO (branch each therapist turn → oracle → preference pairs → DPO update) vs GRPO (group-relative update over the same generations). Matched hyper-parameters, matched look-ahead K, matched min-conversation-length filter.
- **Loop.** Each iteration regenerates all conversations from the current policy; those same conversations are the evaluation set.
- **Evaluation.** Full-conversation scores on eight metrics: the five questionnaire rubrics **Q1+Q2, WAI-SR, CSQ-8, MI-SAT, MITI**, plus **PCT** (patient change-talk), **MICI** (MI-inconsistent therapist behaviour, lower is better), and the MITI ratios **R:Q / %CR / %MICO**. Every conversation is now graded twice — once by the training oracle, once by a held-out judge.

Arm	Iterations trained	Iterations scored	Status
PTO K=0	1–10	base + 1–10	complete, both judges
GRPO K=0	1–10	base + 1–10	complete, both judges
PTO K=5	1–5	base + 1–4	paused (iter-5 adapter unscored)
GRPO K=5	1	base + 1	paused

RESULTS · 1

All metrics across 10 iterations (K = 0)



Mean over the 96 personas at each iteration; shaded band = 95% CI. Iteration 0 = untrained base model. MICI is inverted in meaning (lower = fewer MI-inconsistent therapist behaviours).

Endpoint numbers (K = 0)

Two endpoints reported for each arm: the matched final iteration (10) and each arm's own best iteration on its training rubric (Q1+Q2).

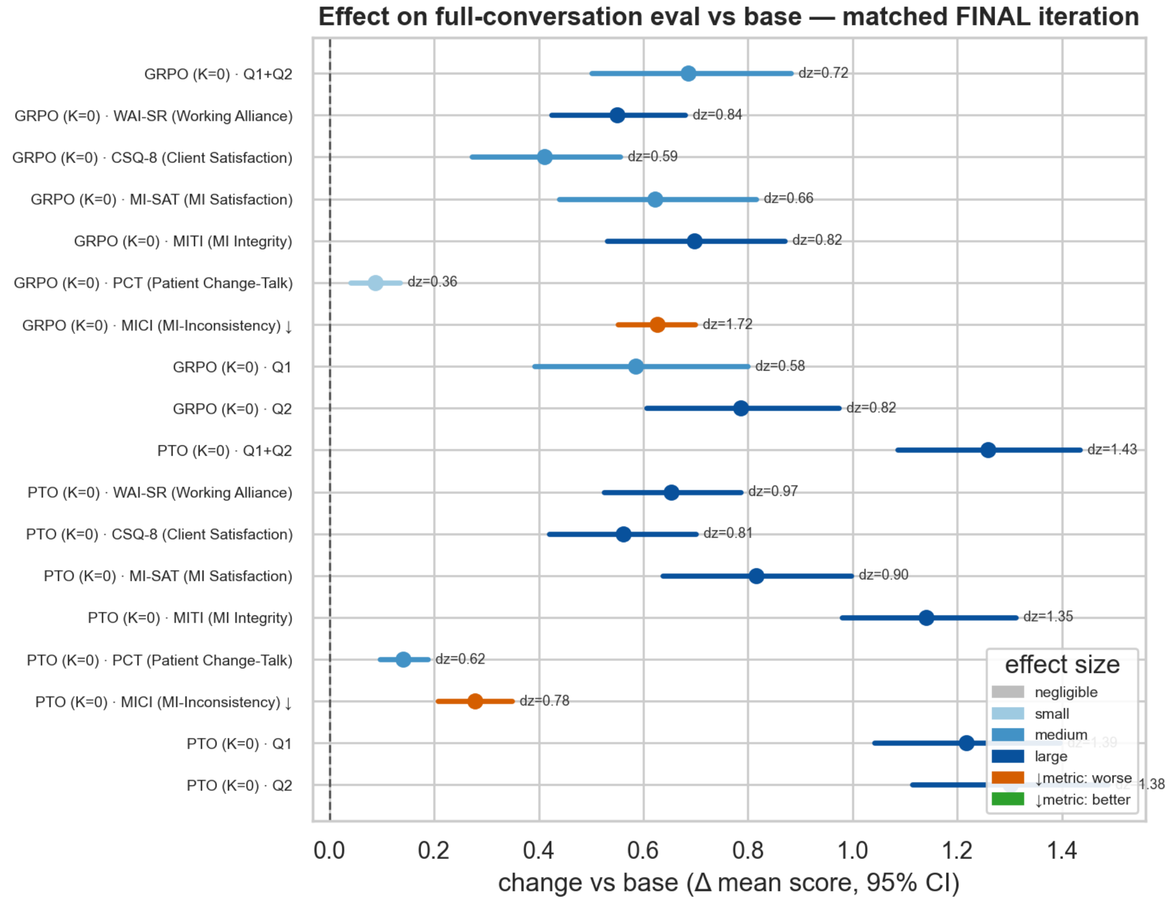
endpoint	arm	iter	Q1+Q2	WAI-SR	CSQ-8	MI-SAT	MITI	PCT	MICI ↓	R:Q	%CR	%MICO
final	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
final	GRPO (K=0)	10	3.753	3.438	2.773	3.479	3.922	0.574	0.838	1.435	0.414	0.833
best	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
best	GRPO (K=0)	8	4.082	3.374	2.749	3.441	4.234	0.572	0.535	1.043	0.372	0.711

Paired contrasts on the 96 shared personas (PTO − GRPO), Holm-corrected

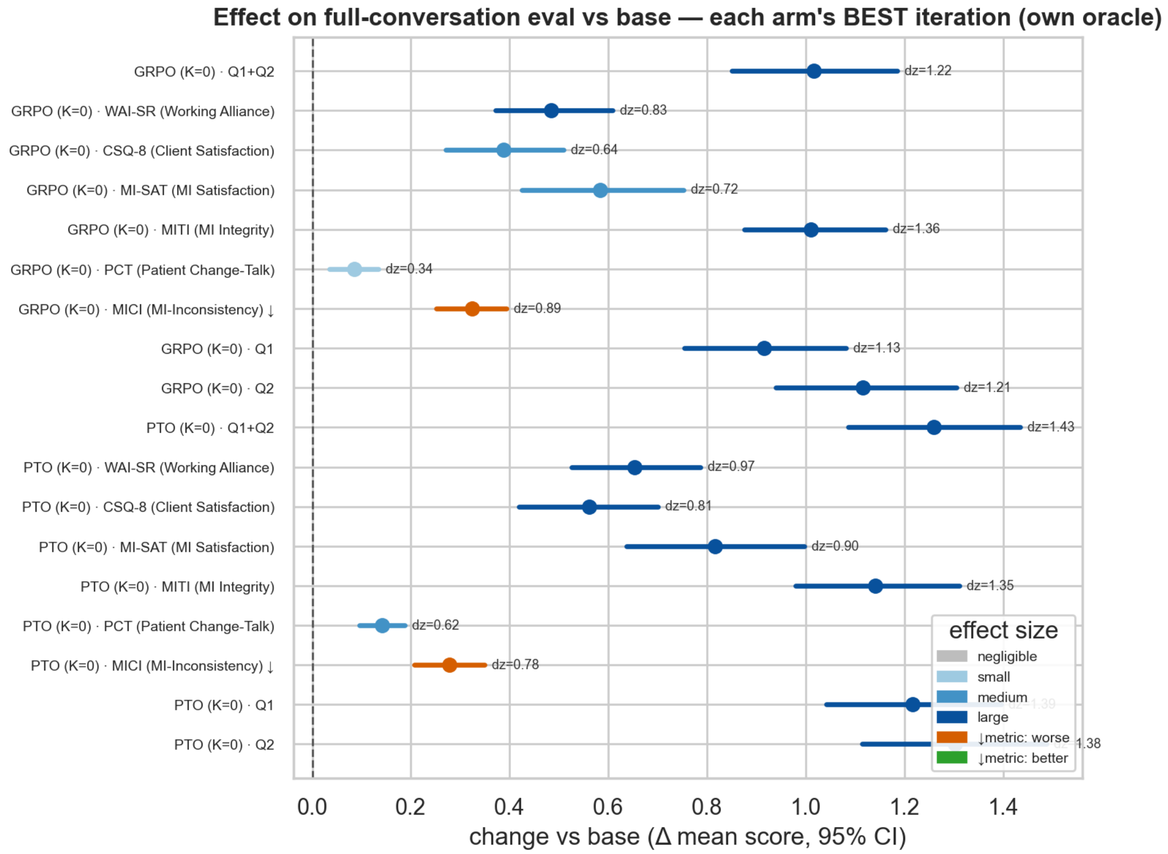
Contrast	Q1+Q2 Δ	dz	p (Holm)
PTO iter-10 vs GRPO iter-10 (matched final)	+0.51	0.73	< .001
PTO iter-10 vs GRPO iter-8 (each at its own best)	+0.18	0.30	.010

Trajectory shape: PTO Q1+Q2 rises 3.00 → 4.26 with its maximum at the last iteration (OLS slope 0.120/iter). GRPO rises 3.07 → 4.08 by iteration 8, then falls to 3.75 by iteration 10 (slope 0.072/iter). Every arm × rubric effect vs base is significant at Holm $p < .001$.

Effect vs the untrained base model (K = 0)



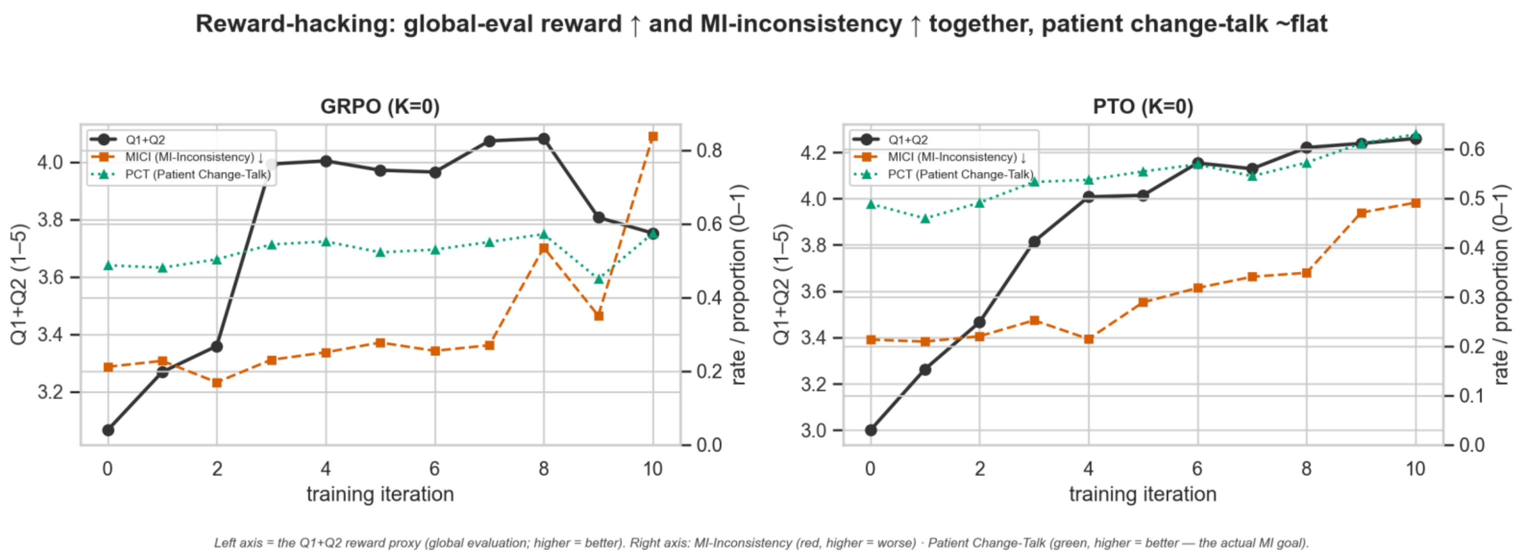
at the matched final iteration (10)



at each arm's own best iteration (PTO 10, GRPO 8)

Paired Cohen's dz vs base over the 96 personas, with 95% CI. Positive = improvement, except MICI where positive = more MI-inconsistent behaviour.

What else moved as the scores rose (K = 0)

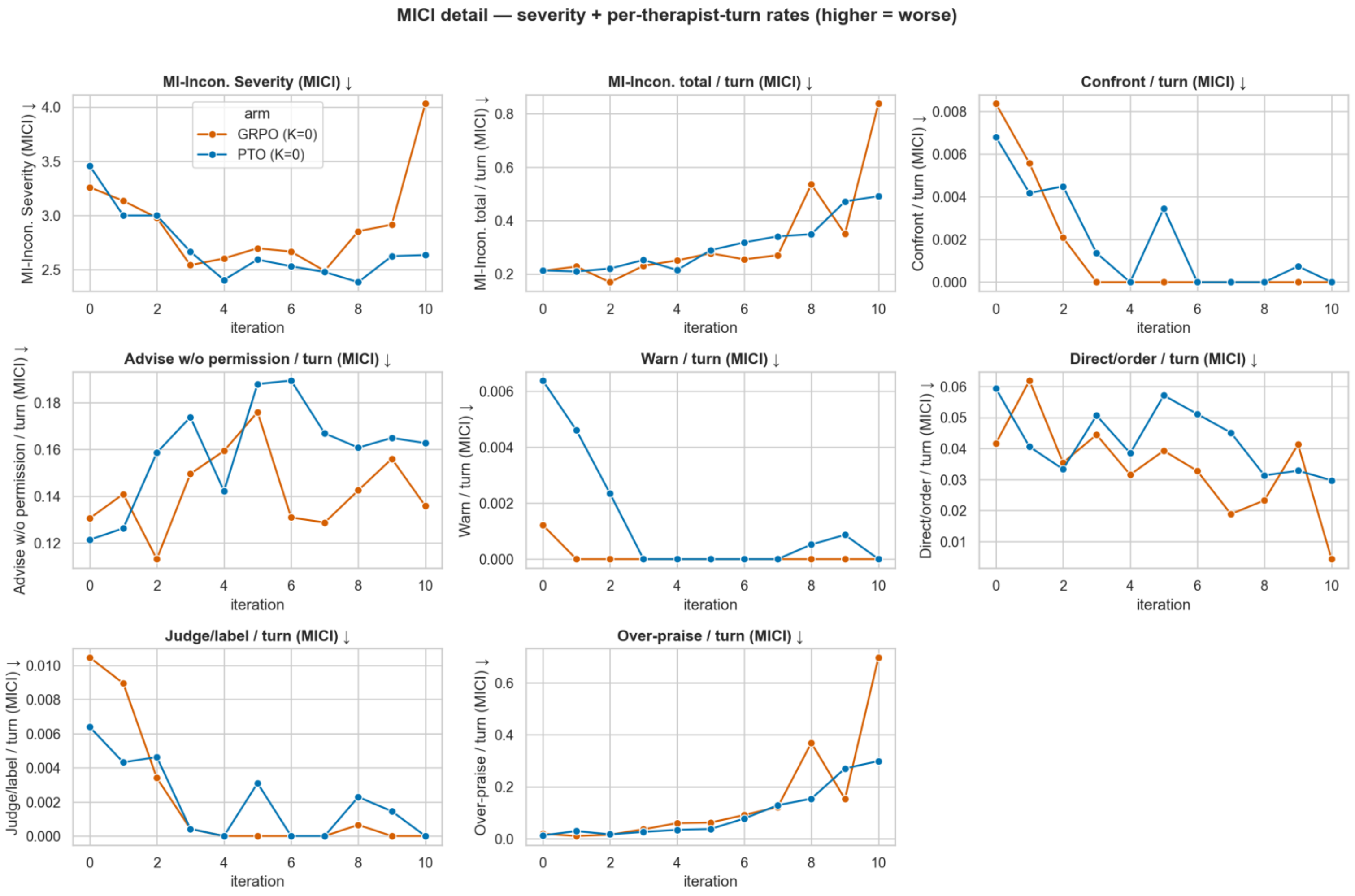


Global rubric score (left axis) vs MI-inconsistent behaviour and patient change-talk (right axis), per iteration

Measured alongside the gains

- **MICI** (MI-inconsistent behaviour) rises 0.21 at base → 0.49 PTO and 0.84 GRPO at iter 10 (GRPO 0.54 at its iter-8 peak).
- GRPO collapses to 0.15 questions per turn by iter 10, against PTO's 0.55; reflection-to-question ratio 1.44 vs 0.75.
- Both arms drift toward affirmation-heavy language, more so in GRPO's late iterations.
- The reward's own composition is part of it: the Q2 items that move most reward therapist self-disclosure, which MI does not prescribe.
- Whether this is really reward-hacking, or just a harsher rubric, is what the held-out judge settles — next section.

MI-inconsistency, decomposed (K = 0)

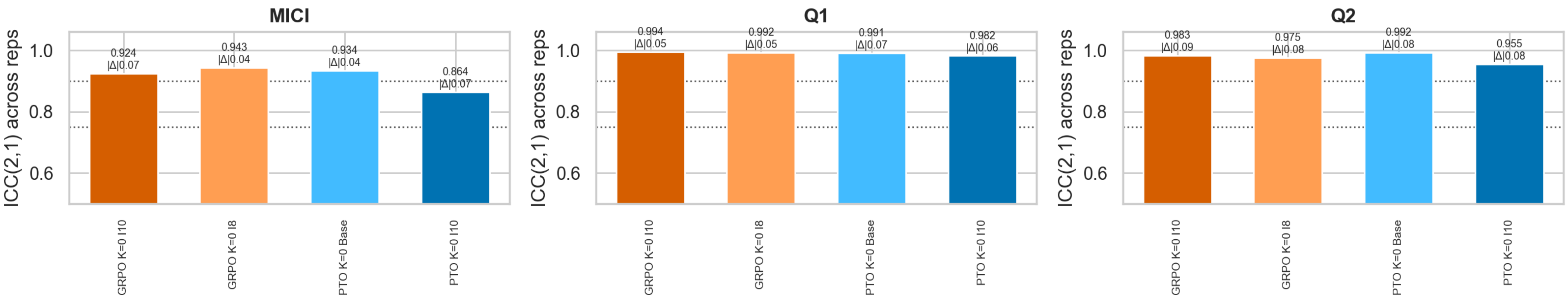


Per-conversation rate of each MI-inconsistent therapist behaviour, by iteration. Lower is better throughout.

MEASUREMENT VALIDITY · 1 · NEW SINCE THE EMAIL

The instrument itself is now measured, not assumed

[EVAL] Oracle repeatability — same conversations re-scored 4× (seeds differ, nothing else); dotted = Koo & Li good (0.75) / excellent (0.90)



ICC(2,1) per model and metric across four independent scorings of the same conversations. Dotted lines: Koo & Li "good" (0.75) and "excellent" (0.90).

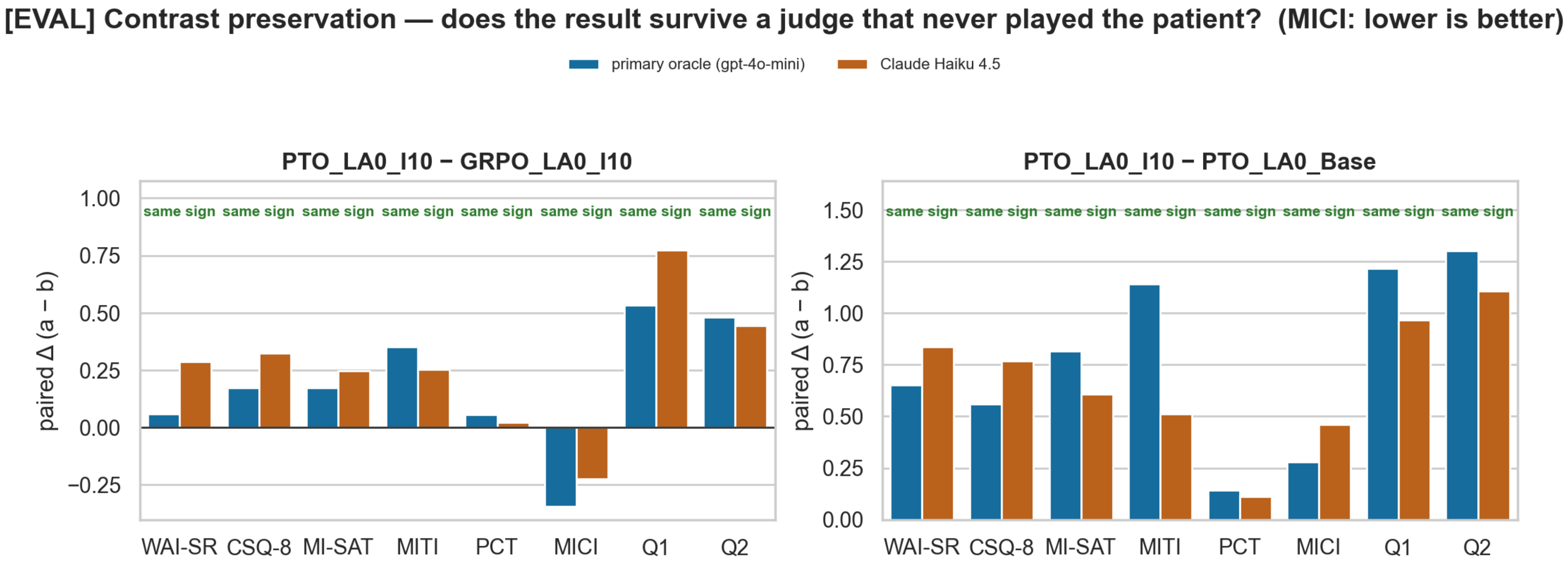
Metric	Training oracle (gpt-4o-mini)	Held-out judge (Haiku 4.5)
Q1	ICC 0.982 – 0.994	ICC 0.951 – 0.978
Q2	ICC 0.955 – 0.992	ICC 0.938 – 0.963
MICI	ICC 0.864 – 0.943	ICC 0.525 – 0.929

Repeated scorings of 4 anchor model states × 96 conversations. Mean |Δ| between repeats 0.04–0.09 — the project's informal "oracle noise ≈ 0.10" was a conservative upper bound, and it shrinks by ~√96 at the arm-mean level everything here reports.

And a second grader over the whole grid

- **Claude Haiku 4.5** — different model family, never played the patient, never touched training — re-scored **22,272 of 22,272 cells** (29 model states × 8 rubrics × 96 conversations).
- Deliberately 1 repeat, not 3: judge noise adds ≈0.01 to a 96-conversation arm mean against ≈0.09 from persona sampling, so breadth beats depth at equal cost.
- Cost **$42** for the sweep plus **$9** for the repeatability repeats, using Anthropic's batch API.

Does the result survive a judge that never played the patient? Yes



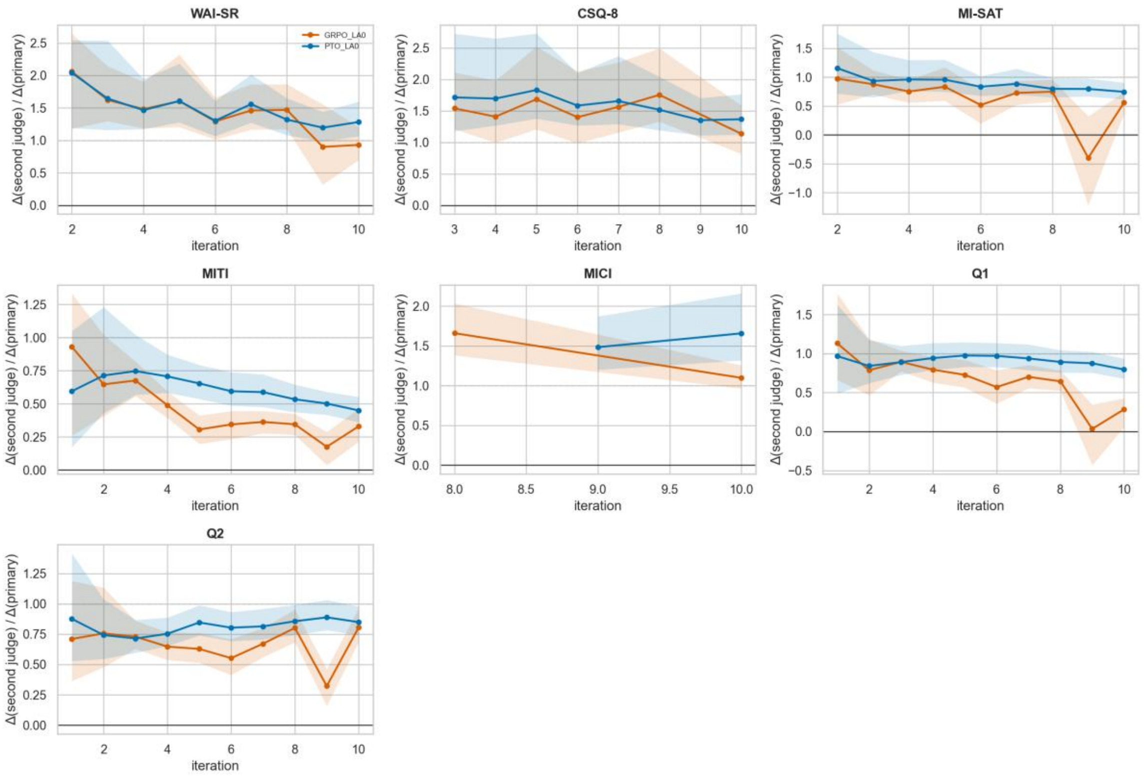
Endpoint contrasts computed separately under each grader. Same-sign bars mean the claim does not depend on the grader also having simulated the patient.

All arm × metric contrasts	|Δ| ≥ 0.10	|Δ| ≥ 0.25	|Δ| ≥ 0.50	Anchor contrasts
88.3% (1,632 / 1,848)	94.1%	97.0%	98.9%	18 / 18

Share of contrasts keeping their sign under the held-out judge. The two graders disagree only about differences too small to claim — and the held-out judge actually widens the headline PTO − GRPO Q1 gap, to +0.77 against the primary's +0.53.

The sharpest evidence for reward-hacking — gain retention

[EVAL] When do the gains stop transferring? Retention vs iteration under Claude Haiku 4.5 (dashed = full transfer; a declining line = progressively fitting the trained-against grader)

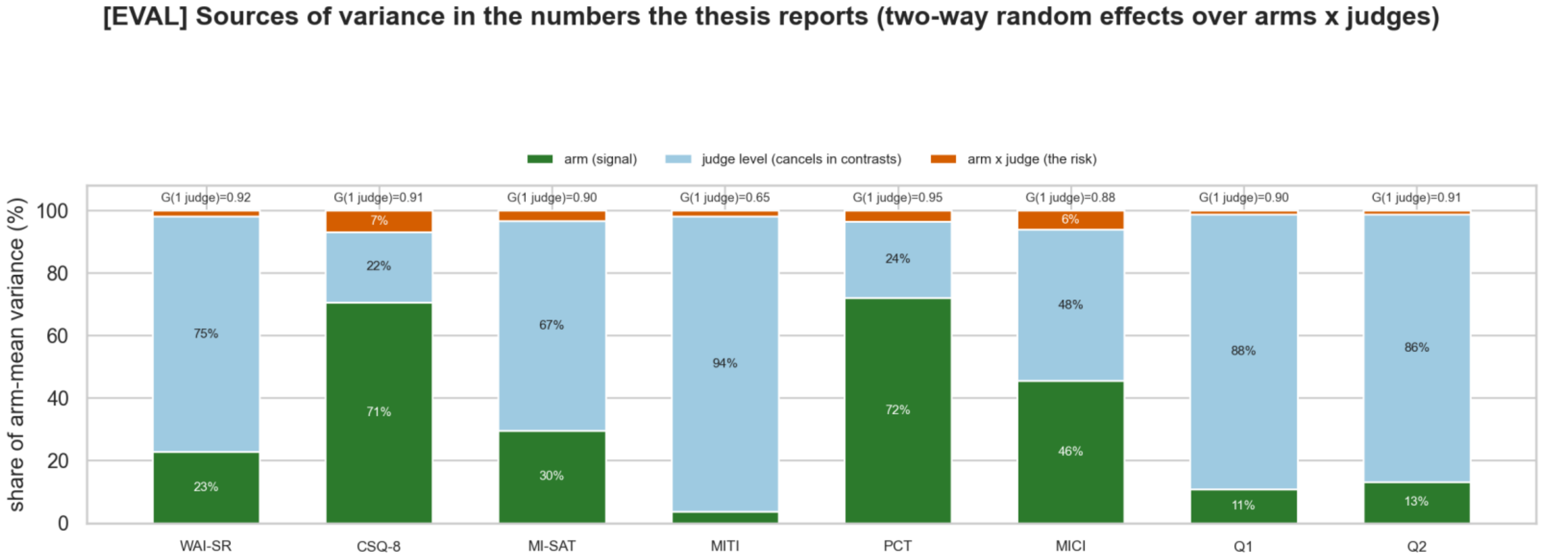


Share of each arm's measured gain that the held-out judge also sees, by iteration.

Read it as a train / test ratio

- The quantity is **Δ under the held-out judge ÷ Δ under the judge that was the training reward**. Near 1.0 = a real gain; falling = a policy fitting the grader it was trained against.
- At iteration 10, Q1 retention is PTO 0.80 [0.68, 0.93] against GRPO 0.28 [0.06, 0.43] — non-overlapping intervals.
- In plain terms: under a grader that never took part in training, GRPO's net 10-iteration Q1 gain is about **0.19 points, not the 0.68** the training oracle credits it with.
- It is an onset, not an endpoint accident. The arms are indistinguishable for three iterations, then PTO holds 0.80–0.98 for the whole run while GRPO decays to 0.28.
- Not a scale artifact: every Q2 retention interval overlaps (0.80–0.85). Only Q1 separates.

Where the two graders do not agree — stated as limitations



Share of arm-mean variance by source. A large grader-level slice is harmless — it cancels in every contrast. The arm × judge slice is the only one that can invalidate a claim, and it is 1.2–6.9% on every metric.

- **MITI is the exception, and it is a thesis limitation.** Only 3.6% of MITI's arm-mean variance is between-arm signal — 94.5% is grader level. Dependability from one judge is 0.65, and it keeps its sign on only 77.5% of contrasts, the worst of the eight. **MITI arm differences are reported as provisional.**

- **MICI agrees weakly across graders (r 0.20–0.55),** and the held-out judge's own MICI repeatability falls exactly where the sycophancy claim lives (0.53 on GRPO at iter 10). So that claim is made at the **contrast level, not as a precise rate** — gain retention is the load-bearing evidence.

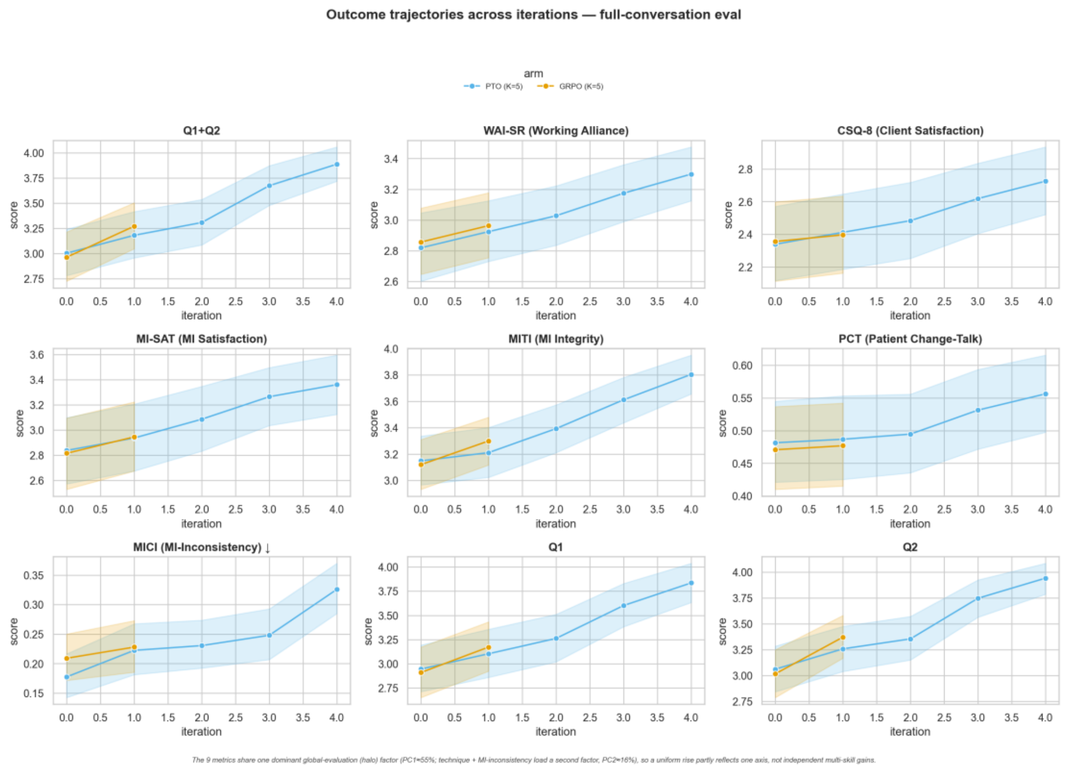
- **Never average the two graders.** The primary was the training reward and the second is held out — train-vs-test, not two raters. The level offset is 1.2–1.7 points and model-dependent, so averaging would silently shrink every effect.

- **Still uncovered: no human MI/MITI-coder validation.** A judge can be perfectly repeatable and consistently wrong, and two LLM judges agreeing does not fix that — the largest remaining validity gap, and it costs time, not budget.

THE OPEN COMPARISON

Look-ahead (K = 5): exactly what exists today

Arm	Trained	Scored	Missing
PTO K=5	iters 1–5	base + iters 1–4	iter-5 eval conversations never generated
GRPO K=5	iter 1	base + iter 1	iters 2–10



K=5 trajectories — 4 iterations for PTO, 1 for GRPO. Read as preliminary.

Numbers so far (PTO K=5 only)

- Q1+Q2 3.00 → 3.89 over 4 iterations (dz 0.88).
- MICI 0.18 → 0.33 over the same 4 iterations.
- K=0 vs K=5, paired at matched early iterations (PTO): no significant difference on any metric (all Holm $p > .5$).
- GRPO K=5 has one trained iteration — not comparable.
- Both K=5 arms were paused on API cost, not on any result.

Against the ICLR paper: there K=5 beat K=0 on Llama-2-7B over 7 iterations. Here the K comparison rests on 4 iterations of one method — underpowered, not negative.

DECISION

Decision 1 — which story do we tell, and where

The three framings are not exclusive, but one of them has to lead: it decides the venue, what still needs running, and how the thesis chapter is shaped.

A · Method story — preference trees vs group-relative optimization under an expensive LLM judge (complete today)

The finding is the stability and generalization gap: GRPO is competitive up to its peak, then overshoots into a grader-specific optimum; PTO sustains gains. Gain retention under a held-out judge is the evidence, and it is unusually clean. Needs nothing further — it is in hand.

B · Look-ahead story — the ICLR lever, extended to both method families (needs the K=5 arms)

The most direct continuation of the paper, and the one Kfir co-authored. Requires finishing at least one K=5 arm; the current K evidence is 4 iterations of one method and shows nothing either way. Needs budget — see Decision 2.

C · MI story — what it takes to train a small model toward genuine MI quality (needs human validation)

The most interesting story to a clinical audience, and the one the added metrics were built for: the questionnaires go up while MI-inconsistent behaviour also goes up. To carry it, the rubrics need a human MI-coder anchor. Needs coder time, not API budget.

My reading: lead with A because it is finished and defensible, fold B in as the look-ahead section if the budget is approved, and keep C as the framing of the thesis chapter rather than of the paper. Happy to be argued out of it.

Decision 2 — what, if anything, to run next

Spent so far: ≈ $300 on OpenAI (training rollouts, look-ahead, oracle scoring) and ≈ $51 on Anthropic (the full second-judge sweep plus repeatability). Cost scales with the number of candidates scored — prompts × G, or branch-points × M — times iterations. Prompt caching is already maxed, so the only lever is call count, not price.

1 · The nearly-free look-ahead point (do regardless)

One generate-only pass with the PTO K=5 iteration-5 adapter that already exists (96 conversations, no training), then score it. Adds a fifth K=5 point for a few dollars.

2 · Resume one K=5 arm, cost-capped (needed for framing B)

Halve the candidate count (M/G 8 → 4) and cap at 5–6 iterations — the curves plateau by iteration 4 anyway. Makes the look-ahead comparison conclusive rather than preliminary. Keep K and the oracle model fixed, since those are the variables under test.

3 · Human MI-coder validation on a sample (needed for framing C)

A trained coder scores a sample of conversations against the same rubrics. Costs time, not budget, and is the only thing that closes the "repeatable but possibly wrong" gap.

4 · A different starting model

Train from an instruction-tuned base rather than the raw base. Changes the starting point substantially and may change how much headroom either method has. Largest cost of the four.

What I'd like to leave with

- **A decision on the lead framing** (A / B / C) — everything else follows from it.
- **A yes or no on resuming one K=5 arm, cost-capped**, and if yes, roughly what budget I should plan against. The free iteration-5 point I will take either way.
- **A view on human coder validation** — whether it is worth arranging, and who could do the coding.
- **Agreement on scope** — what goes in a paper, what stays in the thesis, and whether they are written in parallel.

Already on disk, in case a question turns out to be cheap to answer

- **2,784 scored conversations** across 29 model checkpoints, every one graded on all eight instruments by **both** judges.
- **Per-item decomposition of every instrument**, persona-level heterogeneity splits, and per-candidate generation records from every training iteration — all computed, no further API calls needed.
- **Figures, tables and written summaries regenerate from one command**, under either grader, with seeded confidence intervals.